

OS PRACTICAL

PRACTICAL-8

1. Develop a program using malloc(), calloc(), realloc(), and free(). Monitor memory allocation behavior and identify memory leaks using Valgrind.

Develop a program that demonstrates Copy-on-Write behavior after a fork(). Analyze memory usage before and after modifying data in the child process.

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rishitha@Rishitha: ~/Practical x + v
rishitha@Rishitha:~$
rishitha@Rishitha:~$ mkdir Practical8
rishitha@Rishitha:~$ cd Practical8
rishitha@Rishitha:~/Practical8$ nano memory_alloc.c
rishitha@Rishitha:~/Practical8$ gcc memory_alloc.c -o memory_alloc
rishitha@Rishitha:~/Practical8$ ./memory_alloc
malloc completed
calloc completed
realloc completed
free completed
rishitha@Rishitha:~/Practical8$ valgrind --version
valgrind-3.26.0
rishitha@Rishitha:~/Practical8$ valgrind --leak-check=full ./memory_alloc
==622== Memcheck, a memory error detector
==622== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==622== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==622== Command: ./memory_alloc
==622==
malloc completed
calloc completed
realloc completed
free completed
==622==
==622== HEAP SUMMARY:
==622==    in use at exit: 0 bytes in 0 blocks
==622==    total heap usage: 4 allocs, 4 frees, 1,104 bytes allocated
==622==
==622== All heap blocks were freed -- no leaks are possible
==622==
==622== For lists of detected and suppressed errors, rerun with: -s
==622== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
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rishitha@Rishitha:~/Practical8$ nano cow.c
rishitha@Rishitha:~/Practical8$ gcc cow.c -o cow
rishitha@Rishitha:~/Practical8$ ./cow
Before fork: data[0] = 10
Child before modification: data[0] = 10
Child after modification: data[0] = 20
Parent after child modification: data[0] = 10
rishitha@Rishitha:~/Practical8$
```